



SAFEX SOLUTIONS

Passive Security & Compliance Audit

PASSIVE SECURITY & COMPLIANCE AUDIT REPORT

safexsolutions.com

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INTERNSHIP DOMAIN	Cyber Security
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EXECUTIVE SUMMARY & OBJECTIVE

The objective of this module is to design, simulate, and document an Incident Response (IR) Tabletop exercise for SafeX Solutions. Building upon the Week 1 passive security audit, this exercise models a critical data breach scenario to test defensive capabilities, containment strategies, and response timelines using the NIST SP 800-61 r2 Incident Handling framework.

VULNERABILITY RESEARCH & OSINT INTEGRATION

During the Week 1 passive reconnaissance phase, the following security gaps were identified:

- **Exposed Staging Subdomain:** *dev.safexsolutions.com was found publicly resolvable via Subfinder.*
- **Insecure Architecture:** Lack of IP restrictions, VPN enforcement, or firewall access control lists (ACLs) on staging/development assets.
- **Threat Vector:** Attackers can passively discover the staging footprint and target it for unauthenticated database backups or code leaks.

SIMULATED BREACH SCENARIO: "OPERATION DEVLEAK"

- **Threat Actor Profile:** External opportunistic attacker.
- **Attack Vector:** Scanning and discovery of unhardened staging subdomains (dev.safexsolutions.com).
- **Impact:** High. Unauthorized access to internal database mockups and potential exfiltration of proprietary configurations.

Incident Response Timeline (NIST Framework)

Phase 1: Detection & Analysis (09:00 AM)

- **Event:** Security monitoring tools register an anomaly (high outbound data transfer) on the development subdomain.
- **Action:** Incident handler triggers the playbook and verifies traffic source.

Phase 2: Containment (09:15 AM)

- **Event:** Source of leakage is identified as an unauthenticated database backup directory.
- **Action:** Blue Team implements containment. Option chosen: Network isolation and deployment of access control lists (ACLs).

Phase 3: Eradication & Remediation (09:45 AM)

- **Event:** Malicious connections are severed.

- **Action:** Public DNS records for decommissioned test features are purged, and remaining dev environments are pushed behind a secure VPN gateway.

Phase 4: Post-Incident Activity (10:15 AM)

- **Event:** —
- **Action:** Documentation of lessons learned, firewall policy review, and implementation of automated subdomain monitoring.

PROJECT DOCUMENTATION: INTERACTIVE IR TABLETOP PLAYBOOK

1. What was Built

An interactive Incident Response (IR) Tabletop Playbook simulation tool written in Python. It translates the theoretical tabletop exercise of a subdomain data breach (based on Week 1 OSINT findings) into an executable program.

2. How it Works

The program guides a Blue Team analyst through the critical phases of the NIST Incident Response lifecycle:

- **Identification Phase:** Displays alerts and analyzes firewall logs.
- **Containment Phase:** Pauses and prompts the analyst to make a critical defensive decision.
- **Eradication/Resolution Phase:** Executes the selected choice and outputs the response outcome.

3. How to Run / View It

Prerequisites: Python 3.x installed on Windows, macOS, or Linux.

Execution:

- Open your terminal or Command Prompt.
- Navigate to the folder containing the file: `cd path/to/your/folder`
- Run the script: `python incident_response.py`

MODULE DEMONSTRATION SCREENSHOTS

```
C:\Windows\system32\cmd.exe - python incident_response.py
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Nice computer>cd desktop

C:\Users\Nice computer\Desktop>python incident_response.py
=====
SAFEX SOLUTIONS - INCIDENT RESPONSE TABLETOP PLAYBOOK
=====

[CRITICAL ALERT] Threat Intel: Unauthorized data exfiltration detected!
Target Footprint: dev.safexsolutions.com (Exposed Staging Subdomain)

--- PHASE 1: IDENTIFICATION ---
[+] Analysing Firewall Logs...
[!] Attacker is actively downloading database backups!

--- PHASE 2: CONTAINMENT (Your Action Required) ---
Select Blue Team Defensive Action:
1. Isolate server and enforce strict IP Allow-listing / VPN.
2. Do nothing and monitor.

Enter choice (1 or 2): _
```

```
Enter choice (1 or 2): 1

--- PHASE 3: ERADICATION & RESOLUTION ---
[+] Deploying network isolation...
[+] SUCCESS: Public access revoked. dev.safexsolutions.com is now secured!
[+] Incident Status: RESOLVED.

=====

C:\Users\Nice computer\Desktop>
```

SAFEX SOLUTIONS — WEEK 2 PROGRESS REPORT

Completed Milestones

- **Day 1-2:** Coordinated with Group 33 members to map out module boundaries and prevent duplicate incident scenarios.
- **Day 3:** Conducted threat modeling on the exposed 'dev.safexsolutions.com' subdomain identified in Week 1.
- **Day 4:** Designed and documented the NIST-aligned Tabletop Incident Response timeline.
- **Day 5:** Developed and debugged the interactive Python Playbook simulation script (incident_response.py).
- **Day 6:** Successfully executed the script, took system screenshots of the working playbook, and drafted the final markdown reports. Compiled submission package and prepared the presentation walkthrough.